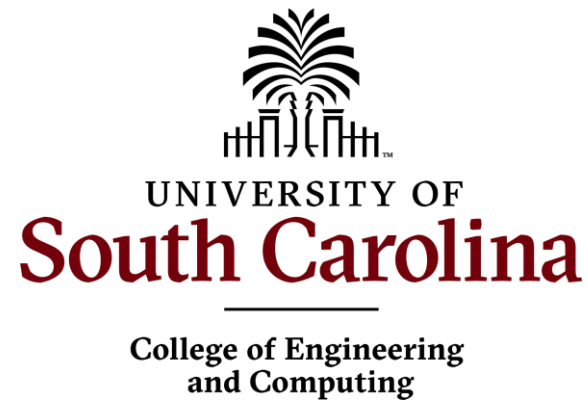


INTRODUCTION TO APPLIED NUMERICAL METHODS

Ordinary Differential Equations: Heun's Method

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EULER'S METHOD

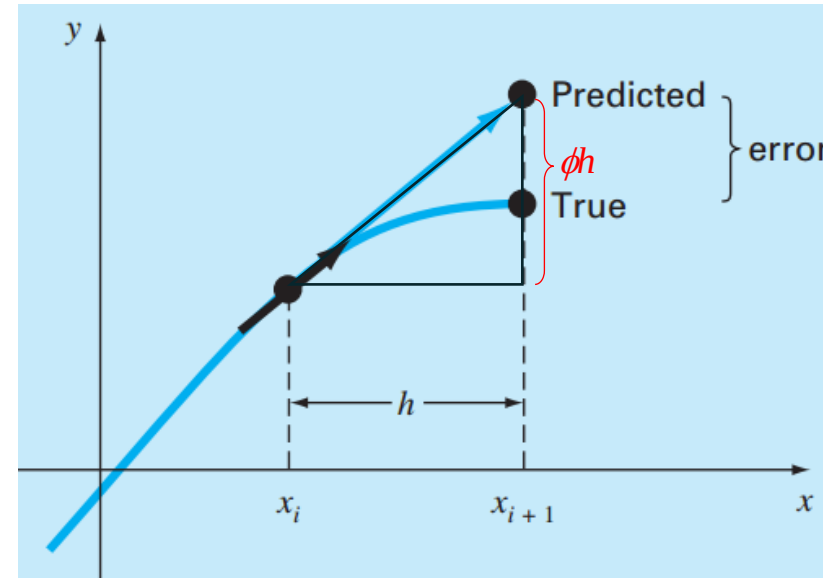
$$y_{i+1} = y_i + \phi h$$

New value = old value + slope $\times$ step size

$$\phi = f(x_i, y_i)$$

ϕ direct estimate of the slope at x_i

$$y_{i+1} = y_i + f(x_i, y_i)h$$



- Prediction based on the first-order Taylor expansion
- A new value of y is predicted using the slope (the first derivative at the original value of x) to extrapolate linearly over the step size h

HEUN'S METHOD

Issue with the Euler's Method

- Derivative at the beginning of the interval is assumed to apply across the entire interval

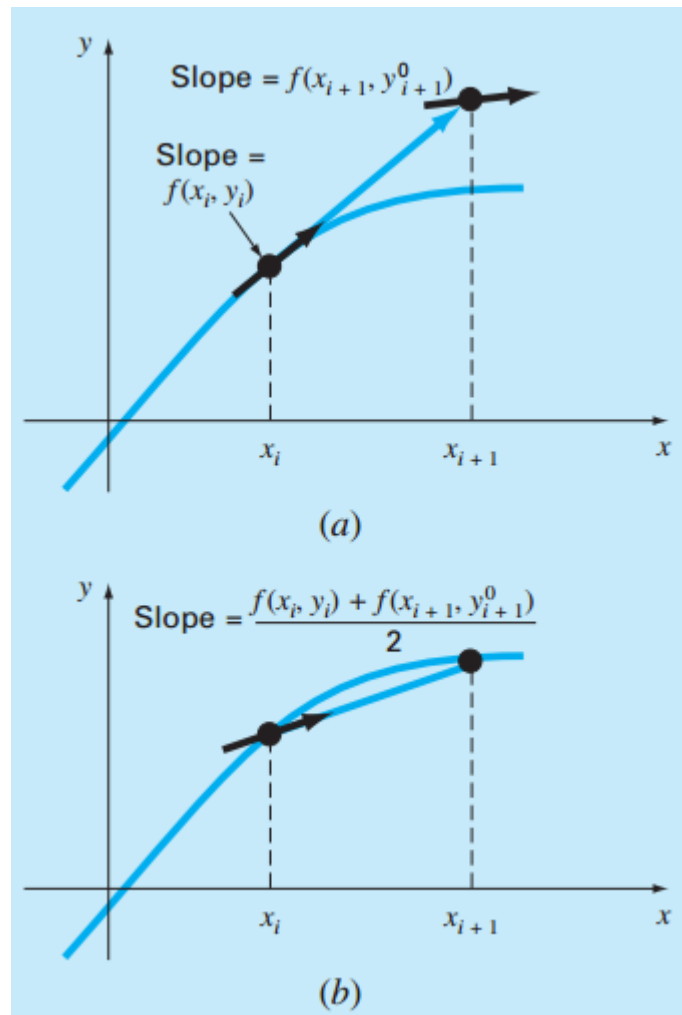


Heun's Method

- Determination of two derivatives for the interval
 - One at the initial point and
 - Another at the end point (**estimated**)
- The two derivatives are then averaged to obtain an improved estimate of the slope for the entire interval

HEUN'S METHOD

Graphic Interpretation



Predictor

$$y_{i+1}^0 = y_i + f(x_i, y_i)h \quad \text{Euler's Method}$$

$$y'_{i+1} = f(x_{i+1}, y_{i+1}^0) \quad \text{Slope estimate at end points } y'_{i+1}$$

Corrector

$$y_{i+1} = y_i + \left[\frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2} \right] h$$

HEUN'S METHOD

$$y_{i+1} = y_i + f(x, y)(x_{i+1} - x_i) = y_i + f(x, y)h$$

1. Initialize x_i and y_i

2. Obtain
 $y'_i = f(x_i, y_i)$

Euler's
Method

3. Compute an
intermediate y^0_{i+1}
 $y^0_{i+1} = y_i + f(x_i, y_i)h$

Predictor

4. Slope **estimate** at
end points y'_{i+1}
 $y'_{i+1} = f(x_{i+1}, y^0_{i+1})$

5. Slope average

$$\bar{y}' = \frac{y'_i + y'_{i+1}}{2} = \frac{f(x_i, y_i) + f(x_{i+1}, y^0_{i+1})}{2}$$

averaging
computing

4. $x_i \leftarrow x_i + \Delta x$
 $y_i \leftarrow y_{i+1}$

4. Compute y_{i+1}

$$y_{i+1} = y_i + \frac{f(x_i, y_i) + f(x_{i+1}, y^0_{i+1})}{2}h$$

Corrector

AN EXAMPLE

Using the Heun's method to numerically integrate

$$\frac{dy}{dx} = f(x, y) = 4e^{0.8x} - 0.5y$$

From $x = 0$ to $x = 4$ with a step size of $h = 1$ with the initial condition $x = 0$ and $y = 2$

Solution:

The exact solution is: $y = \frac{4}{1.3} (e^{0.8x} - e^{-0.5x}) + 2e^{-0.5x}$

1. Initial value: (0, 2)

2. Compute the derivative: $y'_0 = \left. \frac{dy}{dx} \right|_{x=0} = f(0, 2) = 4e^0 - 0.5(2) = 3 \quad \leftarrow y'_i = f(x_i, y_i)$

3. Compute an intermediate value to estimate y at 1.0 (predictor):

$$y_1^0 = 2 + 3(1) = 5 \quad \leftarrow y_{i+1}^0 = y_i + f(x_i, y_i)h$$

4. **Estimate** the slope at the end point:

$$y'_1 = f(x_1, y_1^0) = 4e^{0.8(1)} - 0.5(5) = 6.402164 \quad \leftarrow y'_{i+1} = f(x_{i+1}, y_{i+1}^0)$$

5. Compute the slope average:

$$\bar{y}'_1 = \frac{3 + 6.402164}{2} = 4.701082 \quad \leftarrow y_{i+1} = y_i + \frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2}h$$

AN EXAMPLE

Using the Heun's method to numerically integrate

$$\frac{dy}{dx} = f(x, y) = 4e^{0.8x} - 0.5y$$

From $x = 0$ to $x = 4$ with a step size of $h=1$ with the initial condition $x = 0$ and $y = 2$

Solution:

The exact solution is: $y = \frac{4}{1.3}(e^{0.8x} - e^{-0.5x}) + 2e^{-0.5x}$

The numerical solution:

6. Improve the prediction (corrector):

$$y_1 = 2 + 4.701082(1) = 6.701082 \quad \leftarrow y_{i+1} = y_i + \frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2}h$$

x	y_{true}	y_{Heun}	$ \epsilon_t $ (%)
0	2.0000000	2.0000000	0.00
1	6.1946314	6.7010819	8.18
2	14.8439219	16.3197819	9.94
3	33.6771718	37.1992489	10.46
4	75.3389626	83.3377674	10.62

2ND ORDER RUNGE-KUTTA METHOD

Heun's Method

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2 \right)h$$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + h, y_i + k_1h)$$

$$a_1 = a_2 = 1/2$$

$$p_1 = q_{11} = 1$$

k_1 is the slope at the lower bound and k_2 is the estimated slope at the upper

The Midpoint's Method

$$y_{i+1} = y_i + k_2h$$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1h\right)$$

$$a_1 = 0, a_2 = 1$$

$$p_1 = q_{11} = 1/2$$

k_2 is evaluated at the midpoint of the interval

The Ralston's Method

$$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2 \right)h$$

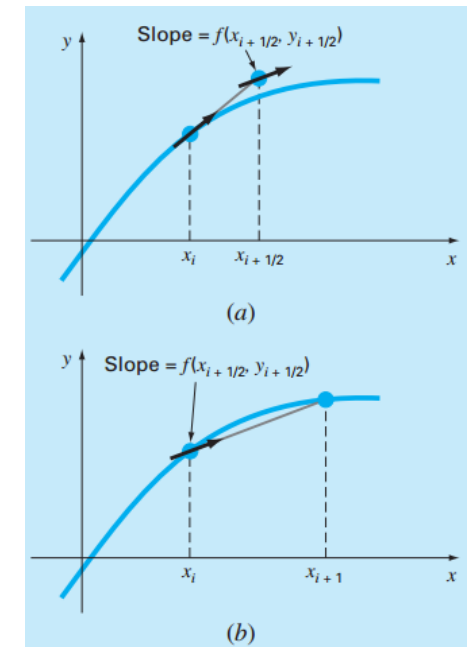
$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1h\right)$$

$$a_1 = 1/3, a_2 = 2/3$$

$$p_1 = q_{11} = 3/4$$

k_2 is evaluated at the $3/4$ location of the interval



AN EXAMPLE

Using the second-order RK schemes (Midpoint and Ralston's) to integrate

$$\frac{dy}{dx} = f(x, y) = -2x^3 + 12x^2 - 20x + 8.5$$

From $x = 0$ and $x = 4$ with a step size of 0.5 with the initial condition $x = 0$ and $y = 1$

Solution: The Midpoint Method

1. Initial value: (0, 1)

2. Compute the derivative: $k_1 = f(x_i, y_i) = f(0, 1) = -2(0)^3 + 12(0)^2 - 20(0) + 8.5 = 8.5$

3. Estimate the slope at the middle point:

$$\left. \begin{aligned} k_2 &= f\left(x_i + \frac{h}{2}, y_i + \frac{k_1}{2}h\right) \\ x_i + \frac{h}{2} &= 0 + 0.25 = 0.25 \end{aligned} \right\} \begin{aligned} k_2 &= -2(0.25)^3 + 12(0.25)^2 - 20(0.25) + 8.5 \\ &= 4.21875 \end{aligned}$$

9 4. Predict y_{i+1} :

$$y_{i+1} = y_i + k_2 h = 1 + 4.21875(0.5) = 3.109375$$

